Robust Noise-Resilient Feature Extraction and 3D-CNN Classification for Hyperspectral Imagery

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ABSTRACT

Hyperspectral image (HSI) classification relies on high-dimensional spectral signatures, often leading to the curse of dimensionality and high computational cost. Most existing methods attempt to mitigate this by selecting dozens of bands; in this paper, we propose a highly efficient framework that uses only three spectral bands. To maximize the information extracted from this minimal spectral subset, we employ a 3D Convolutional Neural Network (3D-CNN) to capture joint spatial-spectral features. We investigate the efficacy of Noise-Adjusted Principal Component Analysis (NAPCA) as a dimensionality reduction tool, comparing it against standard Principal Component Analysis (PCA), Incremental PCA, unsupervised and supervised band selection methods. Furthermore, we implement a spatial data augmentation strategy by integrating rotations and flips to artificially expand the training set and improve model generalization. Experimental results on three real-world HSI datasets show that the proposed approach outperforms other state-of-the-art methods. This approach provides a robust solution for resource-constrained remote sensing applications.

KEYWORDS

HSI classification; Noise adjusted PCA; 3D-CNN

1. Introduction

Hyperspectral imagery (HSI) contains rich spectral information with hundreds of contiguous bands, which can be used to distinguish very similar materials and objects. It has a wide range of applications, such as mineral exploration, precision agriculture, and environmental monitoring [1-4]. Accurate classification is crucial for these applications. Several supervised algorithms have been proposed for HSI classification problems, such as Support Vector Machine (SVM) [5-6], K-nearest neighbor (KNN) classifier [7-8], extreme learning machine (ELM) [9-10], multinomial logistic regression [11-13], random forest (RF) [14-15], convolutional neural network [16-21]. KNN is one of the simplest approaches, which evaluates the Euclidean distance between the training and testing samples and assigns the class label based on the k nearest neighbors. SVM learns an optimal hyperplane that best separates the training samples in a high-dimensional feature space. RF-based methods are tree-based approaches that build an ensemble of diverse decision trees to produce a robust prediction, where each tree is trained on a random subset of training samples and the final class for each test sample is determined by aggregating the predictions of all trees. Multinomial logistic regression is a statistical method for predicting the probability that a pixel belongs to one of several mutually exclusive categories based on its spectral signature.

Recently, deep-learning approaches have attracted increasing attention. In particular, Convolutional Neural Network (CNN) -based methods [22-30] have been applied due to their excellent performance in face recognition, object detection, and video classification. A primary advantage of deep-learning methods is their ability to automatically learn feature representations tailored to the specific problem domain, eliminating the need for manual feature engineering. In general, CNN-based approaches can be divided into three categories: 1D-CNNs, 2D-CNNs, and 3D-CNNs. The 1D-CNN was proposed in [31], and its core idea is to use a single-pixel vector to extract features from the input. The proposed 1D-CNN approach achieved superior performance compared with the SVM approach. To incorporate spatial information from hyperspectral imagery, 2D-CNN approaches are proposed. However, the HSI cube contains rich spectral information. The 3D-CNN [32] extracts the joint spatial-spectral features and achieves superior performance.

Although HSI imagery spans hundreds of spectral bands, many are correlated, leading to the curse of dimensionality. There are typically two types of dimensionality reduction methods [33-35]. The first one is band selection (BS), which selects a subset of bands while still achieving satisfactory classification or detection results. For BS approaches, we may lose important information for classification or detection tasks. The second one is transform-based methods such as principal component analysis (PCA)[36] or linear discriminant analysis (LDA). The main idea is to compress these hundreds of bands into a few principal components that contain the vast majority of the important information. PCA has been widely applied to 3D-CNN methods for HSI classification. However, most approaches focus on performance using tens of spectral channels, which can lead to a high computational burden.

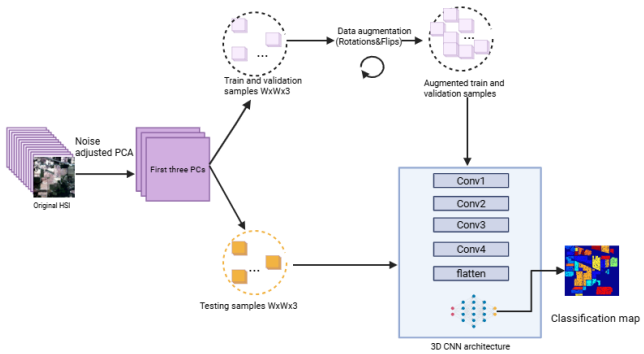

Figure 1: Flow chart of the proposed approach

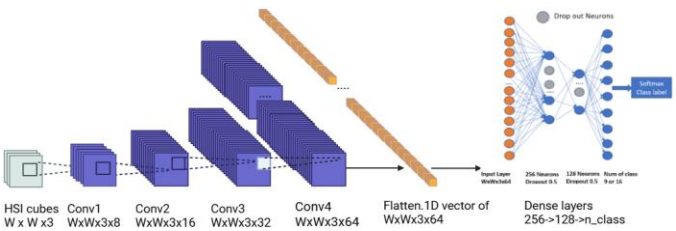

Figure 2. Architecture of the 3D convolutional neural network

Additionally, the first few principal components account for the vast majority of the total variance. By compressing the high dimensional data into a noise-adjusted, low dimensional subspace, it is possible to maintain classification performance while reducing the memory constraints and computation burden.

In this paper, we propose a framework to investigate the performance of 3D-CNN-based HSI classification with extremely reduced input comprising only three spectral bands. Additionally, the traditional PCA may contain noise. To address this, we adopted the noise-adjusted PCA approach to reduce the noise. Finally, we adopted a data augmentation approach to generate robust training samples, ensuring high performance even with minimal spectral inputs

2. Proposed method

The flow chart of the proposed approach is shown in Figure 1. First, the noise was estimated from the hyperspectral cube. Then the NAPCA [37-39] is applied to the original HSI imagery, and the first three principal components are retained. Zero padding is first applied to add a margin of zeros around the three-channel image. A sliding window of n × n is used to extract a cube size of w × w × d, and the label for the cube is taken from the ground truth dataset at the center pixel's coordinates. Second, the constructed cubes are split into training, validation, and testing samples. To ensure model generalization, spatial data augmentation [40] is applied by rotating and flipping the patches during training. Next, the augmented patches are fed into a 3D-CNN that learns the joint spatial-spectral features, and the model is trained. Finally, the test patches are fed into the trained model, which generates the labeled output. The proposed method in this paper is named DA-NAPCA.The architecture of the 3D-CNN is shown in Figure 2. In CNN, a feature map is the output matrix generated when a convolutional filter slides across an input image or previous layer. In the 3D-CNN architecture, a feature map corresponds to a three-dimensional cube. Suppose $\boldsymbol{C_{in}}$ represents the total number of feature cubes as input and m represents the feature cube index. $\mathbf{X}_{\boldsymbol{m}} \in \mathbf{R}^{\boldsymbol{w}\times\boldsymbol{w}\times\boldsymbol{d}}$ represents the $\boldsymbol{m^{th}}$ feature cube of the input data.$\mathbf{W}_{\boldsymbol{m,f}} \in \mathbf{R}^{\boldsymbol{n}\times\boldsymbol{n}\times\boldsymbol{d}}$ represents the 3D kernel filter that belongs to the feature cube and $\boldsymbol{f}$ is the number of kernel filters. For a 3D convolutional layer, the output feature cube of $\mathbf{Y}_{\boldsymbol{f}}$ for the f-th kernel filter is the summation of the convolutions across all input feature cubes.

$$\mathbf{Y}_{\boldsymbol{f}} = \boldsymbol{\sigma}\left(\sum_{\boldsymbol{m}=\mathbf{0}}^{\boldsymbol{C_{in}}-\mathbf{1}}\left(\mathbf{X}_{\boldsymbol{m}} * \mathbf{W}_{\boldsymbol{m,f}}\right) + \boldsymbol{b_f}\right) \quad \textbf{(1)}$$

where * is the 3D convolution operator that calculates the sliding dot product across the height, and $\boldsymbol{b_f}$ is the bias scalar for the f-th filter. The value at the 3D coordinator $(\boldsymbol{i},\boldsymbol{j},\boldsymbol{k})$ is calculated as equation (2), where n, n, and d represent the length, width, and height of the 3D kernels, respectively

$$\mathbf{Y}_{\boldsymbol{f}}^{\boldsymbol{i,j,k}} = \boldsymbol{\sigma}\left(\sum_{\boldsymbol{m}=\mathbf{0}}^{\boldsymbol{C_{in}}-\mathbf{1}}\sum_{\boldsymbol{p}=\mathbf{0}}^{\boldsymbol{n}-\mathbf{1}}\sum_{\boldsymbol{q}=\mathbf{0}}^{\boldsymbol{n}-\mathbf{1}}\sum_{\boldsymbol{r}=\mathbf{0}}^{\boldsymbol{d}-\mathbf{1}}\mathbf{W}_{\boldsymbol{m,f}}^{\boldsymbol{p,q,r}} \cdot \mathbf{X}_{\boldsymbol{m}}^{\boldsymbol{i+p,j+q,k+r}} + \boldsymbol{b_f}\right) \quad \textbf{(2)}$$

where $\boldsymbol{\sigma}$ is the Rectified linear unit (ReLU) activation function, which is shown as equation (3).

$$\boldsymbol{\sigma}(\boldsymbol{x}) = \boldsymbol{max}(\mathbf{0},\boldsymbol{x}) \quad \textbf{(3)}$$

After concatenating all the feature cubes of $\mathbf{Y}_{\boldsymbol{f}}$ into a w × w × d × f tensor Y, the flattened layer unrolls all the values of this tensor Y∈ $\mathbf{R}^{\boldsymbol{w}\times\boldsymbol{w}\times\boldsymbol{d}\times\boldsymbol{f}}$ into a vector z ∈$\mathbf{R}^{\boldsymbol{N}}$. This tensor is then fed into a fully connected neural network with two layers, and the output is calculated as:

$$\mathbf{o} = \boldsymbol{\Theta}^{(3)}\boldsymbol{\sigma}\left(\boldsymbol{\Theta}^{(2)}\boldsymbol{\sigma}\left(\boldsymbol{\Theta}^{(1)}\mathbf{z} + \boldsymbol{b}^{(1)}\right) + \boldsymbol{b}^{(2)}\right) + \boldsymbol{b}^{(3)} \quad \textbf{(4)}$$

where $\boldsymbol{\Theta}^{(i)}$is the weight matrix in the $\boldsymbol{i^{th}}$ dense layer.

Finally, a SoftMax function is applied to determine the output label as in equation (5)

$$\boldsymbol{class\ label} = \boldsymbol{argmax}\left(\frac{\boldsymbol{e^{o_i}}}{\sum_{\boldsymbol{j}=\mathbf{1}}^{\boldsymbol{t}}\boldsymbol{e^{o_j}}}\right) \quad \textbf{(5)}$$

where *t* is the total number of classes.

3. Dataset and Experimental Setup

To evaluate the effectiveness of the proposed approach, we compare it with PCA-based approaches, as well as unsupervised and supervised band selection methods, using two benchmark hyperspectral imagery datasets.

3.1. Indian Pines

The Indian Pines dataset was acquired by the Airborne Visible and Infrared Imaging Spectrometer (AVIRIS) over Indiana's Indian Pines in June 1992. The spatial size is 145 x 145, with a spatial resolution of 20 m/pixel, and the 220 spectral bands range from 0.4 to 2.5 µm. There are 16 distinct classes in the original ground truth. The number of samples per class for the Indian Pines dataset is listed in Table 1.

3.2. University of Pavia

The University of Pavia dataset, collected by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor, comprises 115 spectral bands spanning 0.43 to 0.86 µm, with a spatial resolution

Table 1. Number of samples per class for the Indian Pines

Class No.	Class Name	Number of Samples
C1	Alfalfa	46
C2	Corn-no-till	1460
C3	Corn-min-till	834
C4	Corn	237
C5	Grass-pasture	483
C6	Grass-trees	730
C7	Grass-pasture-mowed	28
C8	Hay-windrowed	478
C9	Oats	20
C10	Soybean-no-till	972
C11	Soybean-min-till	2455
C12	Soybean-clean	593
C13	Wheat	205
C14	Woods	1265
C15	Buildings-grass-trees-drives	386
C16	Stone-steel-towers	93t
	Total	10,285

Table 2. Number of samples per class for the University of Pavia

Class No.	Class Name	Number of Samples
C1	Asphalt	6631
C2	Meadows	18649
C3	Gravel	2099
C4	Trees	3064
C5	Painted metal sheets	1345
C6	Bare Soil	5029
C7	Bitumen	1330
C8	Self-Blocking Bricks	3682
C9	Shadows	947
	Total	42,776

of 610 x 340 pixels. The spatial resolution is 1.3 m/pixel. The number of samples per class for this dataset is listed in Table 2.

4. Experiments

4.1. Experimental Setup

All experiments are performed on the Google Colab Pro platform using the T4 GPU with high RAM. All experiments are conducted 10 times, and the mean and variance of Overall Accuracy (OA), Average Accuracy (AA), and Kappa Coefficient (K) are reported for Indian Pines, University of Pavia. For all comparison approaches, the training set is 10% of the labeled samples, and the validation set is 10% of the labeled samples. 80% of the total samples are used for testing purposes. The experiments are trained using the Adam optimizer, and the learning rate is set to 0.001. The window sizes for Indian Pines, University of Pavia are set to 11 and 7, respectively. The other parameters are set the same for all comparison approaches to ensure a fair comparison. For the Indian Pines, the noisy bands are removed. More specifically, for Indian Pines, the bands 104-108, 150-163, and 220 are removed.

4.2. Comparison methods

Two hyperspectral image datasets including Indian Pines and University of Pavia are used in the experiment. For comparison, several state-of-the-art dimensionality reduction methods are used. The comparison approaches include using RGB, an unsupervised band grouping via clustering approach (CBBS) [41], PCA, Incremental PCA [42], two supervised band selection approaches, including SFS-band [43] and mRMR [44], NAPCA using Gaussian noise, NAPCA using salt-and-pepper noise, and NAPCA using the HSI cube. The difference among these NAPCA variants is that different methods are used to estimate the noise. Gaussian noise with mean 0 and variance 50 was used. Additionally, salt-and-pepper noise was applied to the NAPCA. Another approach is to use the HSI cube to estimate the noise.

4.3. Experimental results and discussion

Table 3, the mean and variance of the proposed and other comparison methods for the OA, AA, and Kappa coefficient are reported for the Indian Pines dataset. It can be concluded that the proposed approach achieves the best performance across all three performance metrics. The PCA-based methods significantly outperform band-selection-based methods, such as RGB, CBBS, SFS-Band, and mRMR. The reason may lie in the limitation of band selection, where the approaches disregard meaningful spectral bands by selecting only a few bands. However, PCA compresses the spectral features, allowing the 3D-CNN to extract those features. Another observation is that, with data augmentation, the OA increases by over 3% relative to the NAPCA. This is expected because Indian Pines is a small dataset with limited training samples. Data augmentation techniques, such as flipping and rotating, can help overcome overfitting.

Table 4 summarizes the classification results for the different dimensionality reduction methods and the proposed DA-NAPCA framework on the University of Pavia dataset. It is evident that the DA-NAPCA and its variant performs the best among the baseline dimensionality reduction techniques. Specifically, the DA-NAPCA approach achieved an OA of 97.9%, which is a significant improvement over unsupervised band selection, such as CBBS, and over supervised band selection, such as mRMR. The DA-NAPCA approach improved the overall accuracy by approximately 4% relative to the PCA and by more than 7% compared to the standard RGB band selection. For the University of Pavia dataset, the impact of data augmentation is less pronounced than in Indian Pines due to its larger training sample size and more balanced class distribution. Nevertheless, the technique still yields a 0.8% improvement in OA. This shows that data augmentation techniques could further increase OA.

The effects of the window size are also investigated for the two datasets for the proposed approach. As shown in Figures 3 (a) and (b), OA increases with larger window sizes across all datasets, and then remains stable as we further increase the window size. With smaller windows, the 3D-CNN architecture cannot capture spatial textures, shapes, and other features. As the window size increase, increases the 3D-CNN can capture richer spatial context thereby increasing OA. However, as the window size increases, the OA remains stable. This may be because a further increase in window size introduces redundant information or mixed labels from adjacent classes, and therefore degrading classification performance.

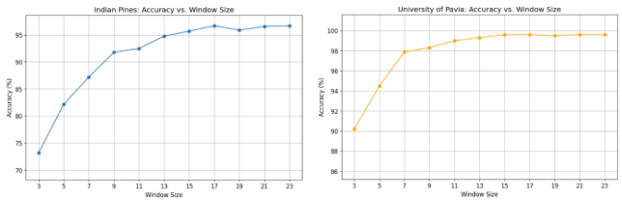

Figure 3. (a) OA vs Window Size for Indian Pines. (b) OA vs Window Size for University of Pavia.

Table 3. Mean and Variance for Overall accuracy (OA), Average accuracy (AA), and Kappa coefficient (K) for Indian Pines.

Method	OA (%)	AA (%)	K × **100**
RGB	89.6±0.37	93.17±0.23	88.52±0.45
CBBS	86.3±0.72	87.2±1.288	84.72±0.93
PCA	93.29±0.11	96.55±0.14	92.53±0.14
Incremental PCA	93.87±0.09	96.66±0.04	93.17±0.12
SFS-Band	77.78±0.42	74.45±3.49	75.08±0.5
mRMR	86.13±3.45	89.64±2.57	84.5±4.39
NAPCA (Gaussian)	96.08±0.01	98.22±0.008	95.64±0.013
NAPCA (salt-and-pepper)	96.084±0.07	98.218±0.025	95.64±0.087
NAPCA	96.64±0.03	98.36±0.01	96.26±0.03
DA-NAPCA	97.27±0.019	98.68±0.018	96.96±0.024

Table4. Mean and Variance for Overall accuracy (OA), Average accuracy (AA), and Kappa coefficient (K) for University of Pavia dataset.

Method	OA (%)	AA (%)	K × **100**
RGB	89.69±0.37	93.17±0.23	88.52±0.45
CBBS	86.3±0.72	87.2±1.288	84.72±0.93
PCA	93.29±0.11	96.55±0.14	92.53±0.14
Incremental PCA	93.87±0.09	96.66±0.04	93.17±0.12
SFS-Band	77.78±0.42	74.45±3.49	75.08±0.5
mRMR	86.13±3.45	89.64±2.57	84.5±4.39
NAPCA (Gaussian)	96.08±0.01	98.22±0.008	95.64±0.013
NAPCA (salt-and-pepper)	96.084±0.07	98.218±0.025	95.64±0.087
NAPCA	96.64±0.03	98.36±0.01	96.26±0.03
DA-NAPCA	97.27±0.019	98.68±0.018	96.96±0.024

5. Conclusion

In this paper, we propose an efficient 3D-CNN framework for hyperspectral imagery classification. We have demonstrated that it is possible to maintain high accuracy with only three principal components. We have validated the effectiveness of noise-adjusted principal component analysis. The experimental results show that the proposed approach could outperform the standard PCA, Incremental PCA, and various unsupervised and supervised band selection methods. Furthermore, integrating spatial data augmentation could address sample-size limitations and mitigate the risk of overfitting. Experimental results on three real-world HSI datasets show that the proposed approach outperforms other state-of-the-art methods.

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